ENHANCED PROFFESSIONAL STUDENT LOGBOOK

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ADMISSION NUMBER: BIT/2024/74164
SCHOOL: COMPUTING AND INFORMATICS
PROGRAMME: BIT4107- MOBILE APPLICATION DEVELOPMENT
ACADEMIC YEAR: MAY/AUGUST 2026
SEMESTER: YEAR 3 SEM 2
LECTURE: MR NYORO MICHAEL
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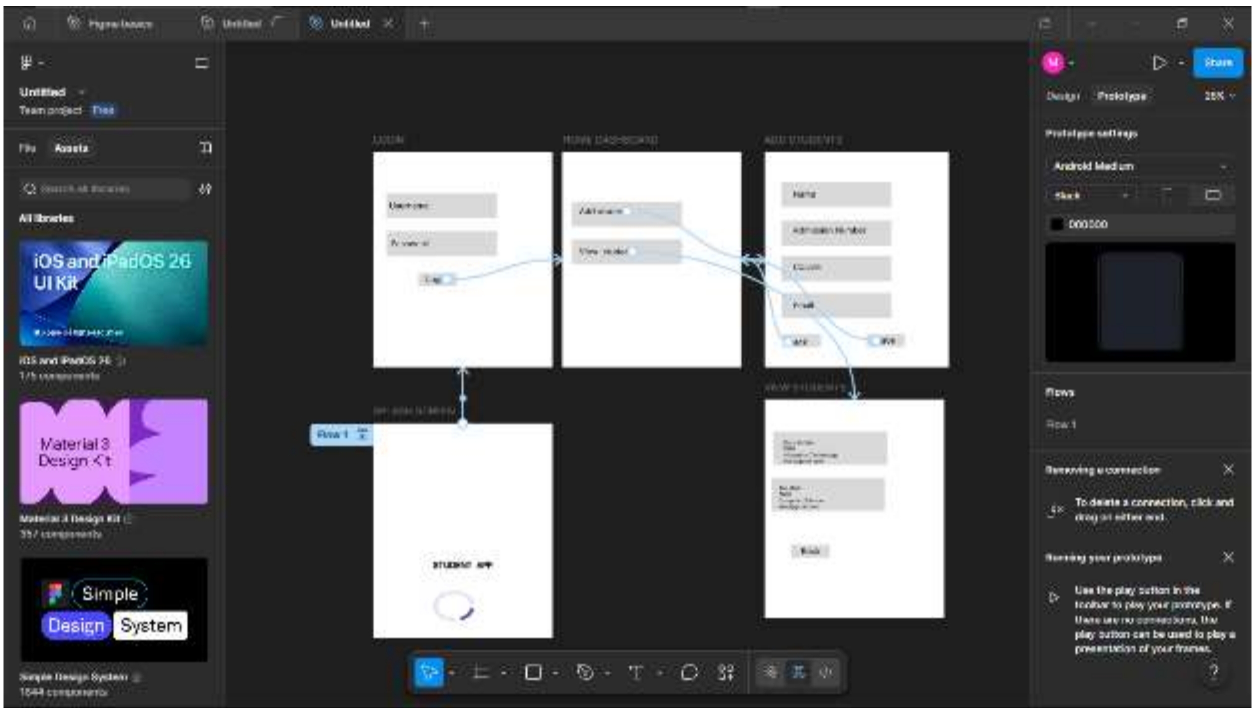
Week 1–3: Flutter Development, Programming Frameworks, and User Interface Design

Figure 1:Figma Wireframes and Prototypes



SYSTEM IMPLEMENTATION(Emulator)

Figure 2:Splash Screen

Figure 3:Login Screen

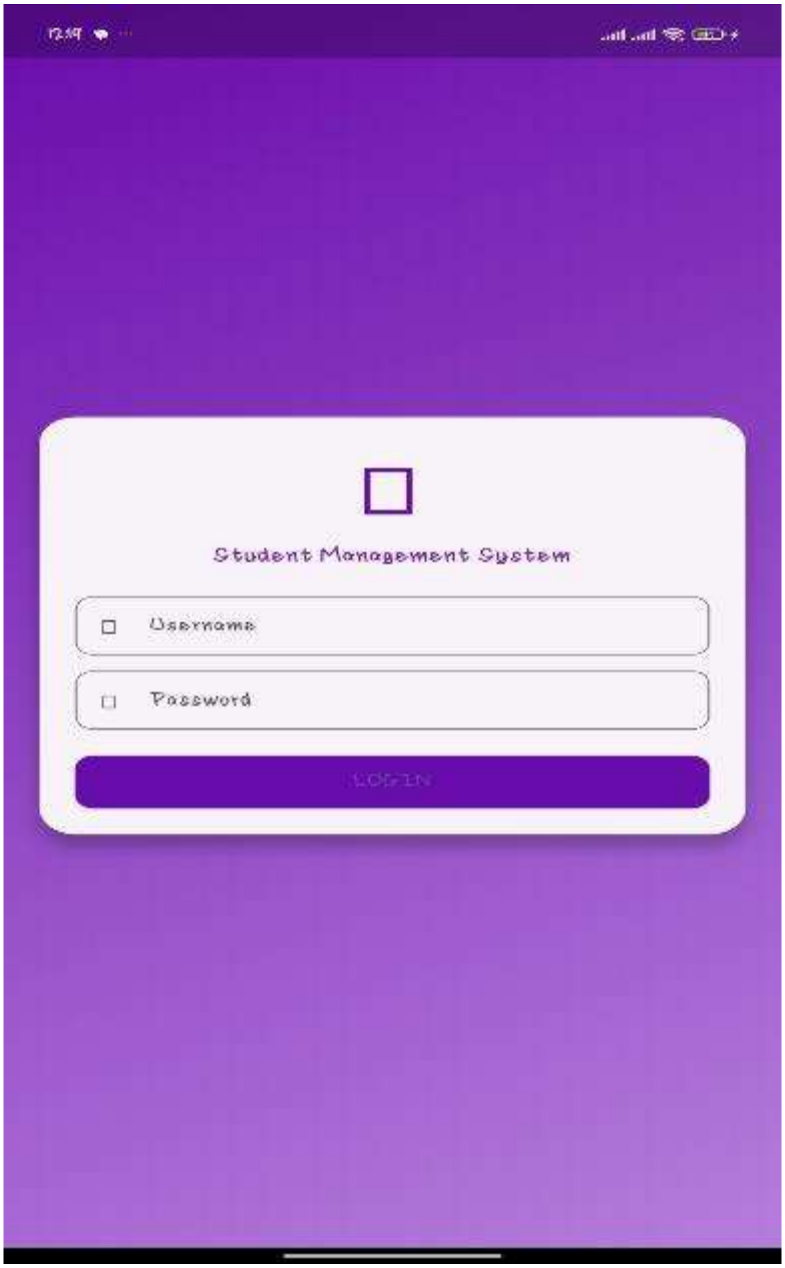
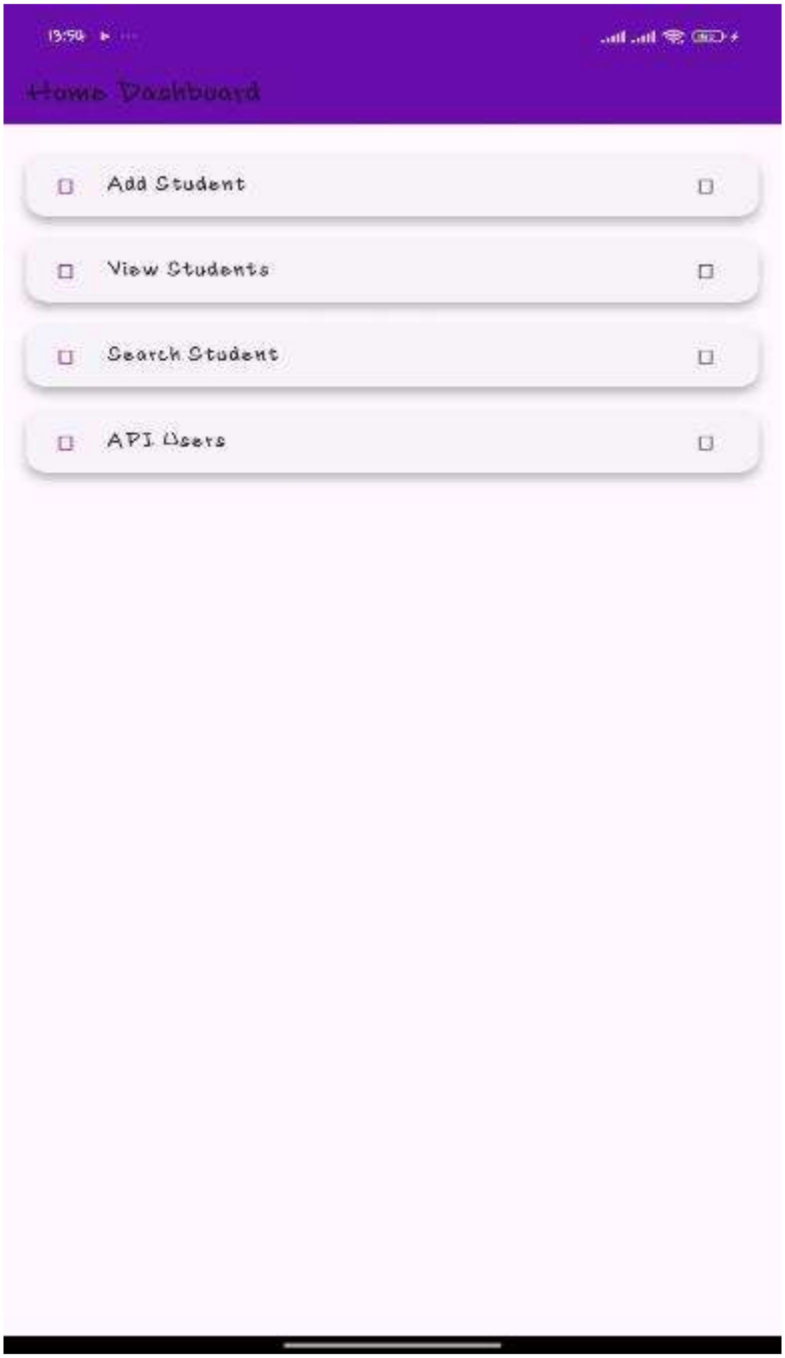


Figure 4:Home Dashboard Screen



Tools Used

- Flutter SDK
- Dart Programming Language
- Visual Studio Code (VS Code)
- Android Emulator
- Figma
- Material Design Widgets

Personal Reflection

These weeks introduced me to Flutter development, application navigation, and user interface design. Creating wireframes and implementing screens improved my understanding of planning, coding, and building user-friendly mobile applications.

Week 4: Data Management

Figure 5:Add Student Screen

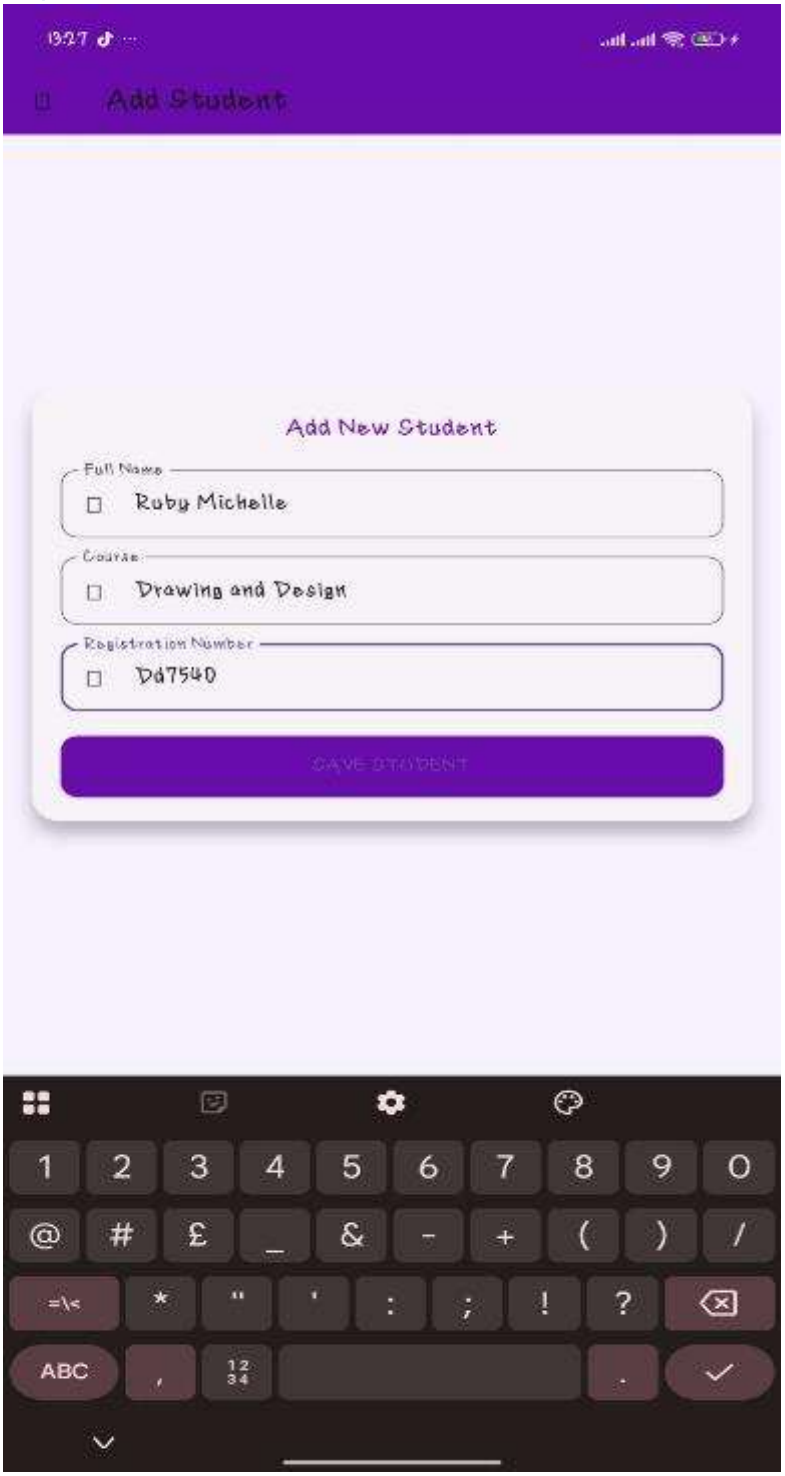


Figure 6:View Student Screen

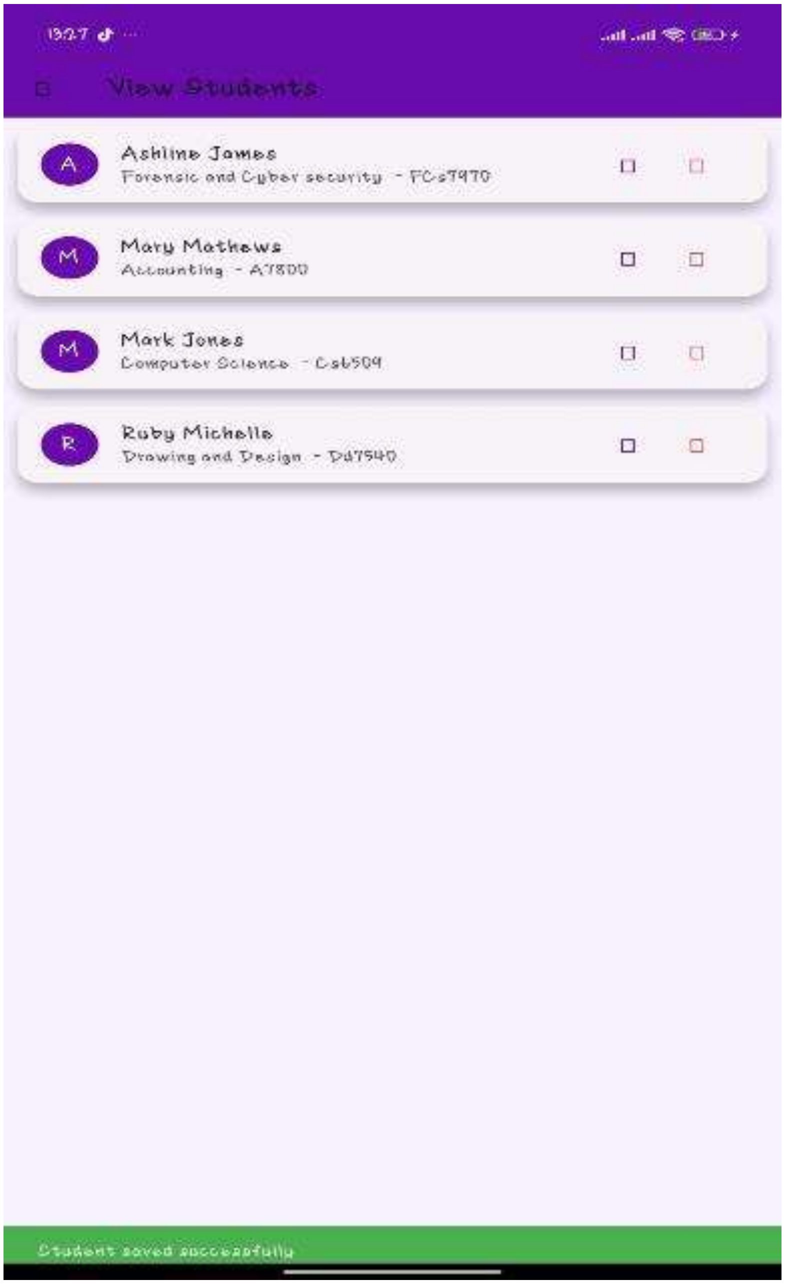
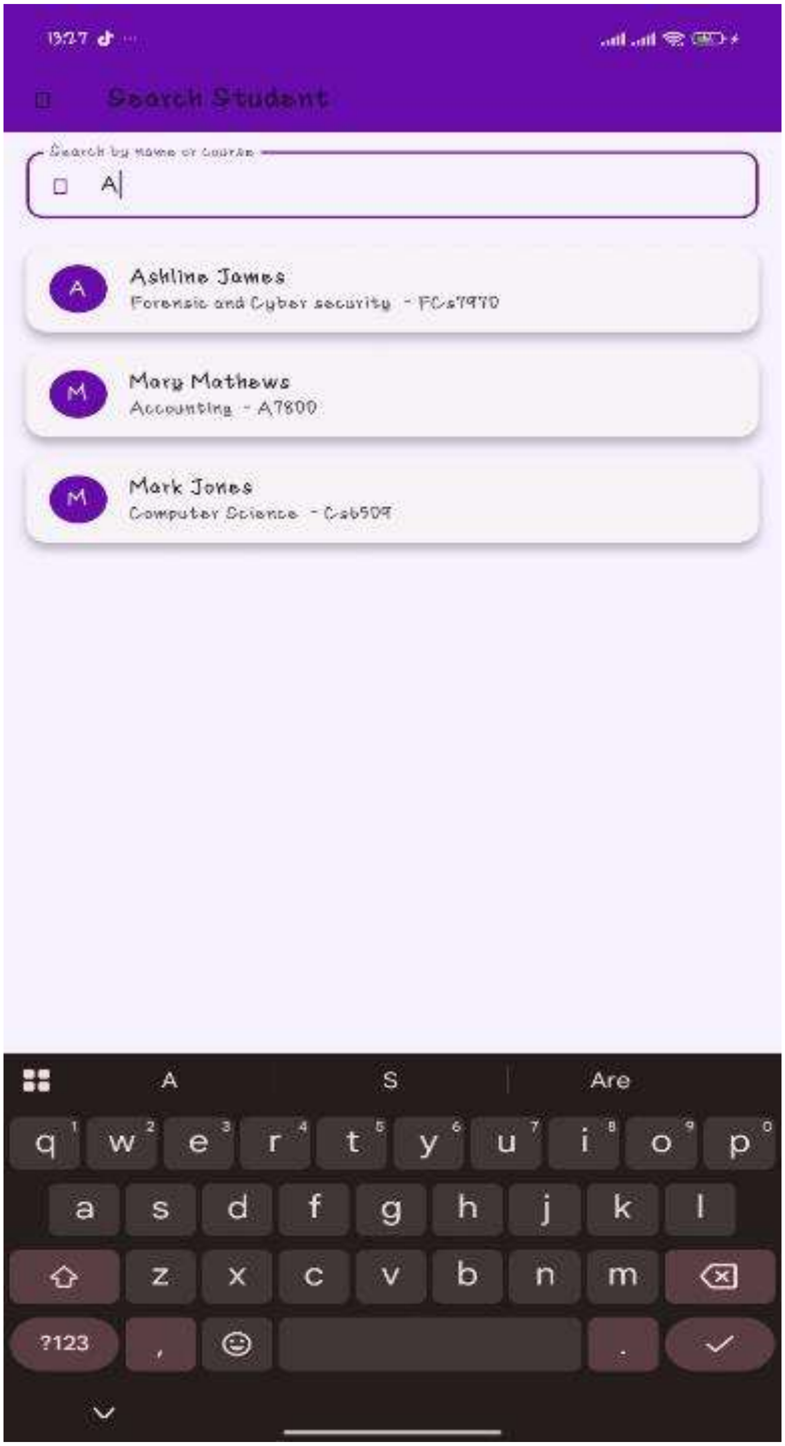


Figure 7:Edit Student Detail Screen

Figure 8:Search Student Screen



Tools Used

- Flutter Framework
- Dart Programming Language

- SQLite Database
- SQFlite Package
- Visual Studio Code
- Android Emulator

Personal Reflection

This week strengthened my understanding of data management. I learned how to add, update, delete, and search student records using SQLite, improving my practical database management skills in Flutter.

Week 5–6: Networking and API Integration

Figure 9:Fetched users from public REST API

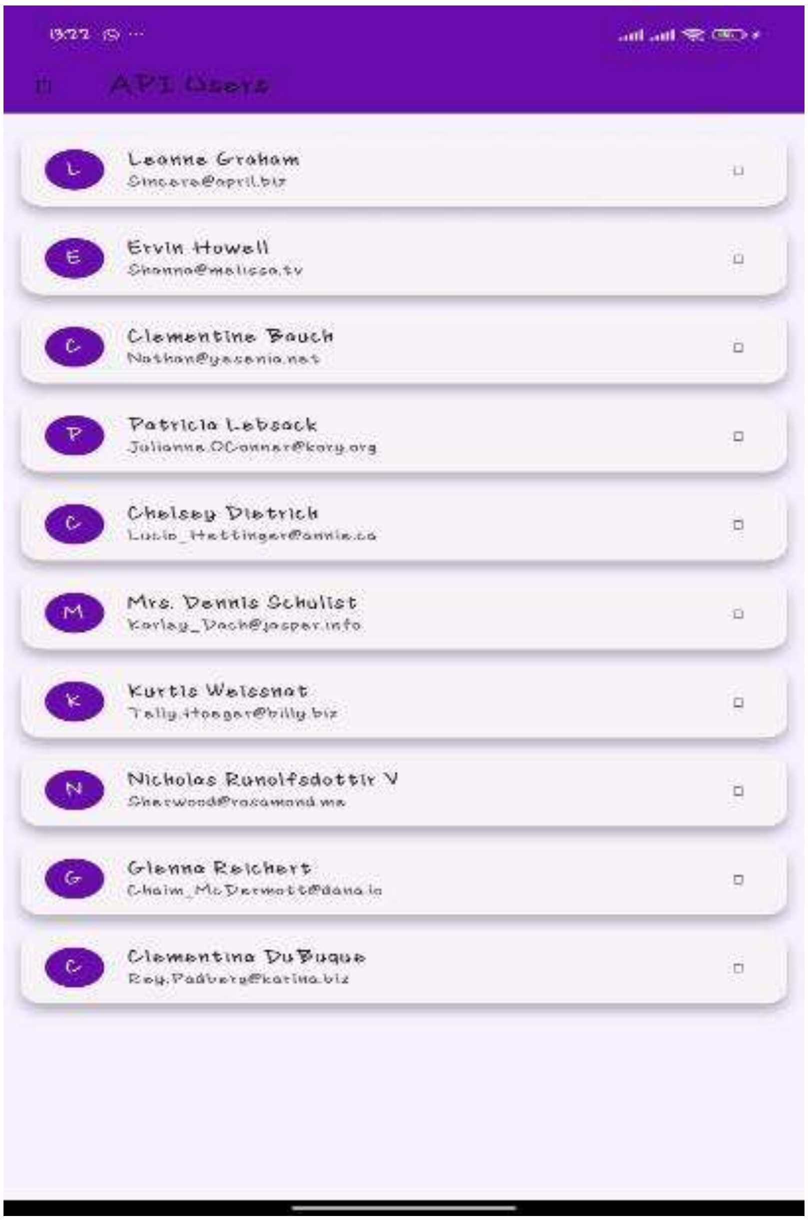
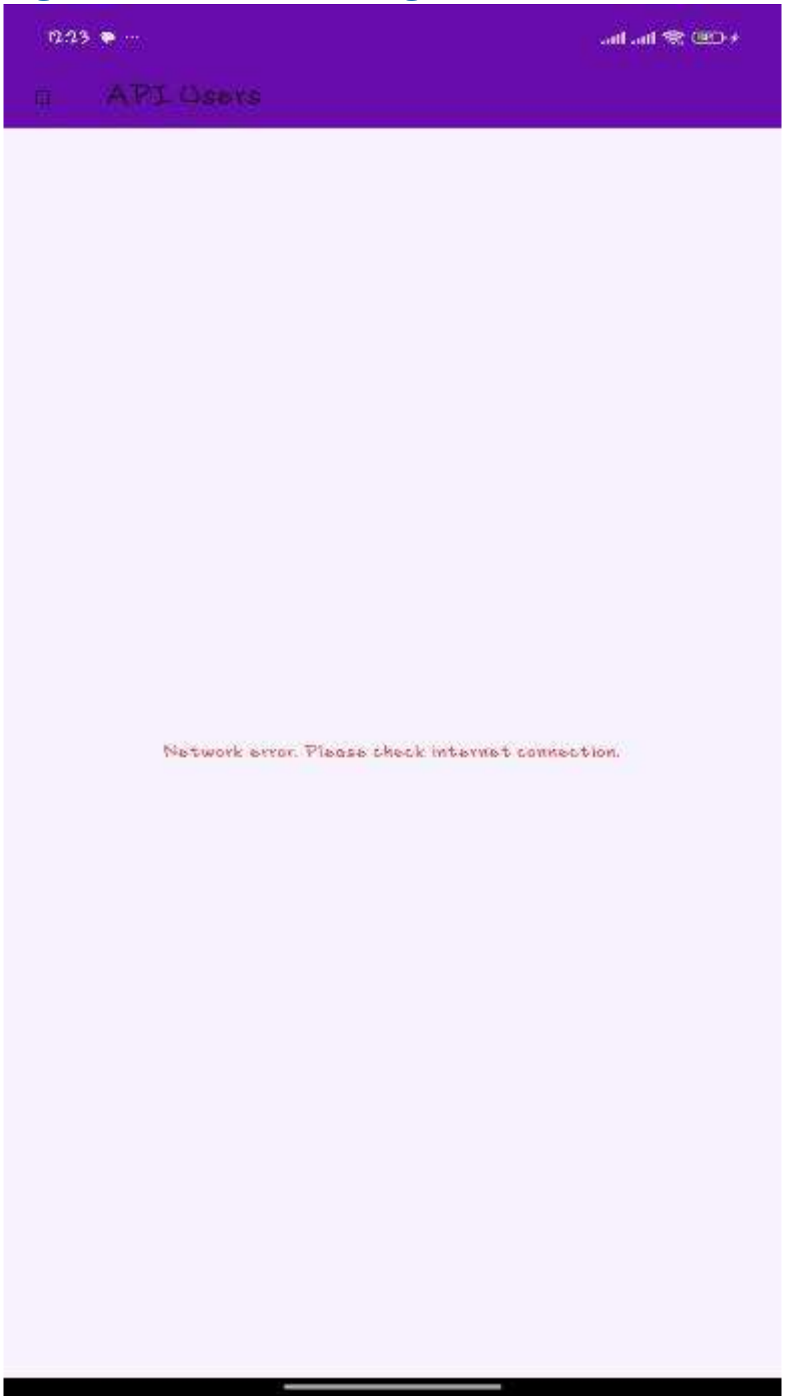


Figure 10:Error handling



Why network is required

The application requires internet connection because it retrieves user data from a remote REST API. Without internet access, the app cannot communicate with the server or display updated records.

Tools Used

- Flutter Framework
- Dart Programming Language
- REST API
- HTTP Package
- JSON Data Format
- Visual Studio Code
- Android Emulator

Personal Reflection

Working with APIs improved my understanding of networking in mobile applications. I learned how to retrieve JSON data from a public API and display it within the application

Week 7: Database Storage and Database Integration

Figure 11:Attendance Record

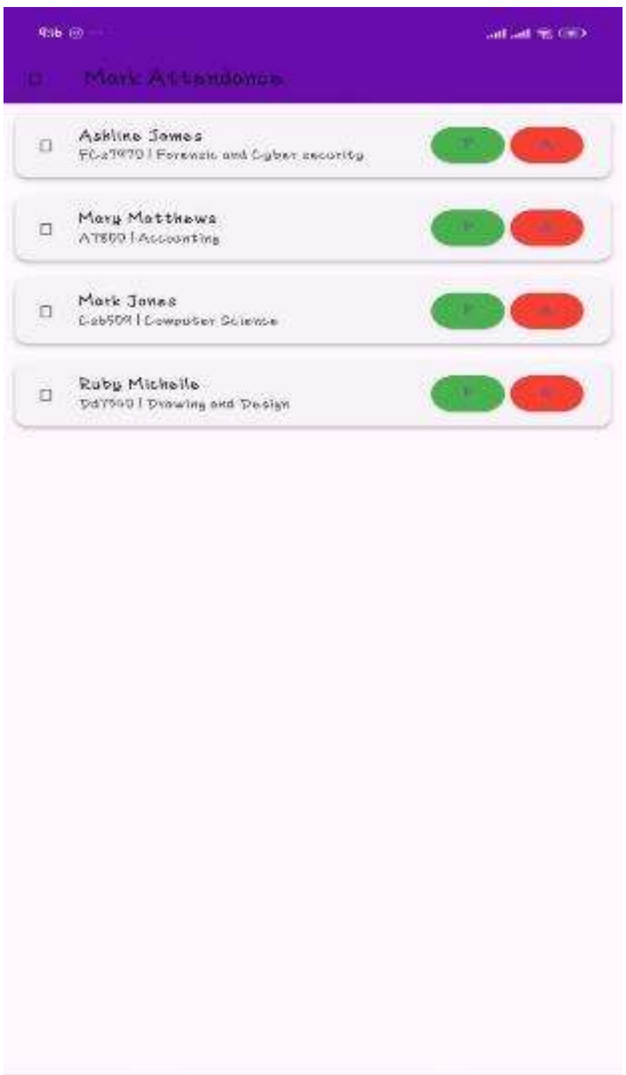
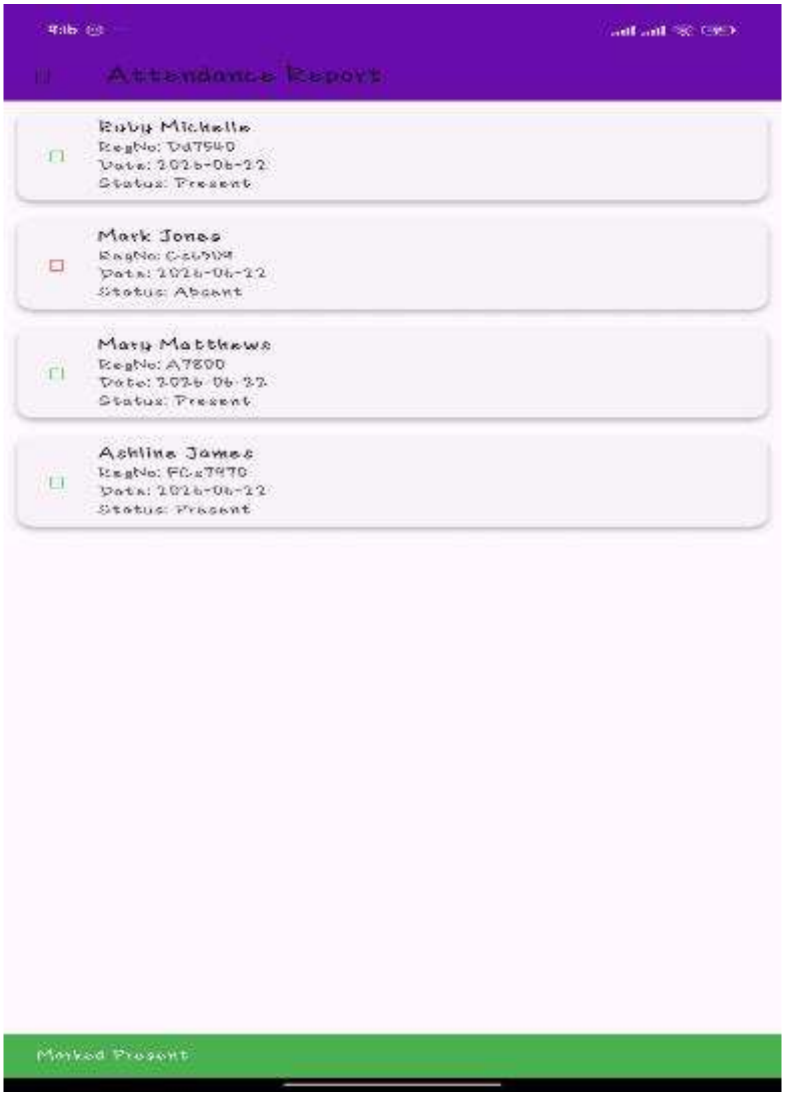


Figure 12:Attendance Report



Tools Used

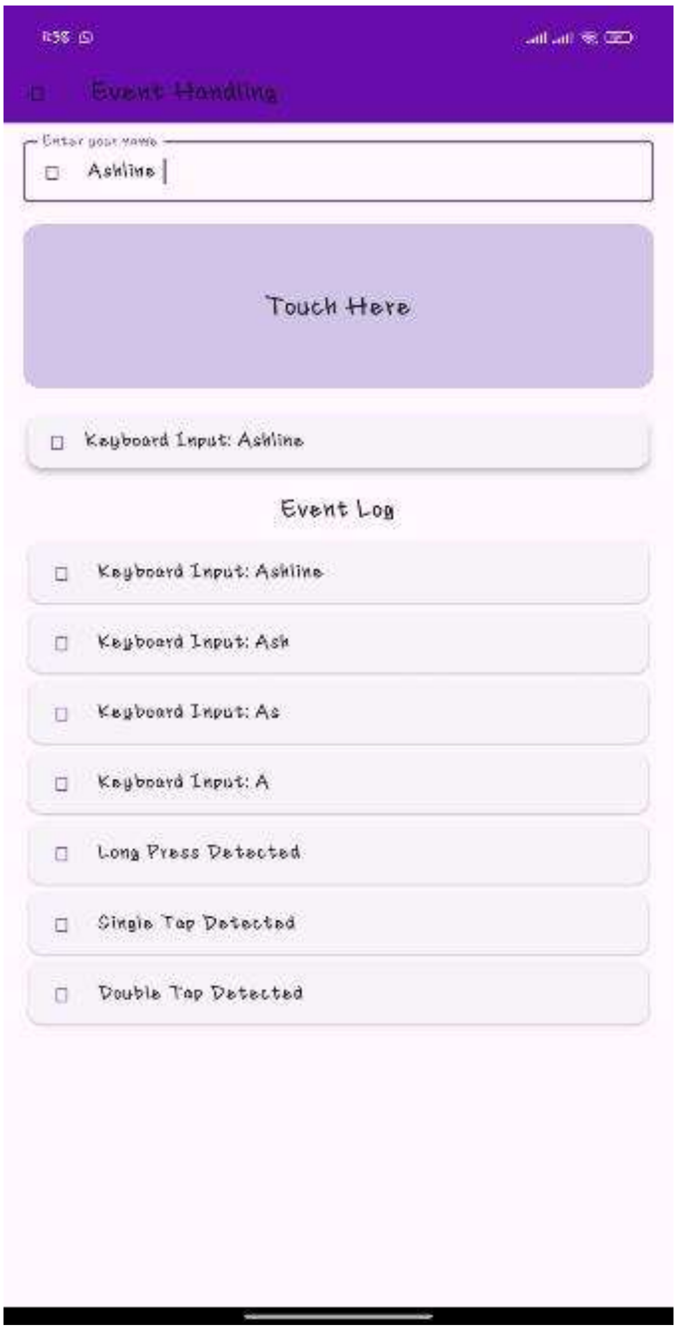
- Flutter Framework
- Dart Programming Language
- SQLite Database
- SQFlite Package
- Visual Studio Code
- Android Emulator
- GitHub

Personal Reflection

Developing the attendance management feature enhanced my database and application development skills. I learned how to manage attendance records, generate reports, and organize project files using GitHub.

Week 8: Event Handling and Object-Oriented Programming

Figure 13:Event handling screen demonstrating keyboard input ,touch gestures and event logging



Tools Used

- Flutter Framework
- Dart Programming Language
- Visual Studio Code
- Android Emulator
- GestureDetector Widget
- TextField Widget

Personal Reflection

This week improved my understanding of event handling and object-oriented programming. I learned to capture keyboard input, detect touch gestures, log events, and organize functionality into separate classes and methods.